Cisco Packet Tracer Project

Step-by-step guide: PC1 → PC2 Request Timed Out; PC1 → PC3 Destination Host Unreachable

1. Project Objective

Build a network using one Cisco 2911 router, one Cisco 2960 switch, and three PCs. The intended tests are: (1) PC1 cannot ping PC2 and shows Request timed out; (2) PC1 cannot reach PC3 and shows Destination host unreachable.

2. Devices and Cable Connections

Place these devices in Packet Tracer: 1 × Router 2911, 1 × Switch 2960, and 3 × PCs. Use Copper Straight-Through cables.

Device	Interface	Connect to
Router	GigabitEthernet0/0	Switch Fa0/1
PC1 (left)	FastEthernet0	Switch Fa0/2
PC2 (middle)	FastEthernet0	Switch Fa0/4
PC3 (right)	FastEthernet0	Switch Fa0/3

3. Configure IP Addresses

On each PC, open Desktop → IP Configuration and enter the following settings:

Device	IP Address	Subnet Mask	Default Gateway
PC1	192.168.10.2	255.255.255.0	192.168.10.1
PC2	192.168.20.2	255.255.255.0	192.168.20.1
PC3	192.168.30.2	255.255.255.0	Leave blank

Note: Leave PC3's gateway blank for this version. PC3 remains physically connected, but the router will not have a VLAN 30 subinterface or a route to its subnet.

4. Configure the Switch

Open Switch → CLI. Enter the commands below. This assigns each PC's switch port to its VLAN and configures the router-facing port as a trunk.

```
enable
configure terminal

vlan 10
exit
vlan 20
exit
vlan 30
exit

interface fa0/2
 switchport mode access
 switchport access vlan 10
```

```
 no shutdown
exit

interface fa0/4
 switchport mode access
 switchport access vlan 20
 no shutdown
exit

interface fa0/3
 switchport mode access
 switchport access vlan 30
 no shutdown
exit

interface fa0/1
 switchport mode trunk
 no shutdown
exit

end
write memory
```

5. Configure the Router

Open Router → CLI. Configure router-on-a-stick subinterfaces only for VLAN 10 and VLAN 20. Do not create a VLAN 30 subinterface for this project version.

```
enable
configure terminal

interface gigabitEthernet0/0
 no shutdown
exit

interface gigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
exit

interface gigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
exit
```

6. Configure an Extended ACL to Block PC1 → PC2 Ping

The ACL silently drops ICMP echo requests from PC1 to PC2. Create the ACL before applying it to the router subinterface.

```
ip access-list extended BLOCK-PC2
 deny icmp host 192.168.10.2 host 192.168.20.2 echo
 permit ip any any
exit

interface gigabitEthernet0/0.10
 ip access-group BLOCK-PC2 in
exit

end
write memory
```

7. Test from PC1

On PC1, open Desktop → Command Prompt.

Test A — PC1 to PC2

```
ping 192.168.20.2
```

Expected result: each ping attempt should show Request timed out.

Test B — PC1 to PC3

```
ping 192.168.30.2
```

Expected result, if no other route to 192.168.30.0/24 exists: Reply from 192.168.10.1: Destination host unreachable.

8. Verification Commands

On the router:

```
show ip interface brief
show access-lists
show ip route
```

On the switch:

```
show vlan brief
show interfaces trunk
```

9. Troubleshooting and Important Notes

• Confirm the router-to-switch cable is connected to Router GigabitEthernet0/0 and Switch Fa0/1, as listed above.
• Confirm the switch ports match the physical topology: PC1 on Fa0/2, PC2 on Fa0/4, PC3 on Fa0/3.
• If PC1 → PC2 returns “Destination host unreachable” instead of a timeout, verify the router subinterfaces and VLAN trunk are up and the ACL is applied inbound to GigabitEthernet0/0.10.
• If PC1 → PC3 replies successfully, check whether a VLAN 30 subinterface or another route to 192.168.30.0/24 exists; remove it for this project version if appropriate.
• PC3 is physically connected, but this configuration intentionally does not provide routed access to its subnet. The PC3 ping result depends on the router's routing table and any other configured routes.

10. Conclusion

This project demonstrates VLAN segmentation, router subinterfaces, and Extended ACLs. The ACL drops PC1's ping requests to PC2, while the absence of a route to PC3's subnet is intended to produce a destination-unreachable message. Verify both results in Packet Tracer.